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Software Requirements Specification

for

SmartReplenish Android Mobile Application

Version 1.0 approved

Prepared by 1376T

FreshTech Systems

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Revision History

Name	Date	Reason For Changes	Version

1. Declarations

1.1 AI Declaration

We acknowledge the use of Gemini, ChatGPT, and Grammarly solely for grammar correction and enhancement. We have reviewed, edited, and verified all AI-generated suggestions. Every technical requirement, system constraint, and functional specifications are the original work of this group.

1.2 Plagiarism Declaration

Employees are not allowed to copy assessment solutions from another person or from other sources. It is the employees' responsibility to guarantee that their solutions are their own work. Meanwhile, the employees must also ensure that their work is not accessible to others. We hereby declare that:

- We fully understand and agree to the above-mentioned policy.
- We did not copy any materials from others or from other places.
- We did not share our materials with others or upload them to any other places for public access.
- We agree that we will not disclose any information or material from the team project to others or upload it to any other places for public access.
- We agree that our project will receive zero marks if any misalignment with the above mentioned policies is detected.

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2. Introduction

2.1 Purpose

The Purpose of this document is to provide a comprehensive Software Requirements Specification (SRS) for SmartReplenish, an Android-based grocery inventory and replenishment management system. This document specifies the functional and non-functional requirements for FreshTech Systems to implement for ValuPrice Pte Ltd. It serves as the technical baseline for system development and final validation.

2.2 Document Conventions

This document follows the standard IEEE 830-1998 SRS template with specific augmentations to enhance traceability and risk management. Specifically, an Inceptive Requirements Traceability Matrix (Section 6) and a Change Mitigation Strategy (Section 9) have been added to ensure alignment between stakeholder needs and functional specifications from the project's onset.

- Requirement Identifiers: Every requirement is assigned a unique, traceable ID based on its type:
    - FR-XX: Functional requirements in Section 5 (System Features)
    - NFR-PR-XX / NFR-SAF-XX / NFR-SEC-XX / NFR-QA-XX: Nonfunctional requirements
    - BR-XX: Business rules
    - CM-ACR-XX / CM-OI-XX / CM-HRCC-XX: Change mitigation items
    - TR-XXX: Technical/Non-functional Requirements (Structural/Security).
    - IR-XXX: Interpretations derived from the original Project Specification (PS) analysis.
- Priority Levels: Requirement priorities are defined using the MoSCoW prioritisation ranking
- Typography:
    - Bold text is used for defined terms or specific actor names (e.g., Household User).
    - Italics are used for document titles or cross-references.
- Imperatives: The document uses strict imperatives to define obligation levels:

- "Shall" or "Must" denotes an essential requirement.
- "Should" or "Desirable" denotes a non-essential but beneficial feature.

· Traceability: Every FR/NFR/BR in this document maps to its source PS requirement via the iRTM in Section 6 and the consolidated appendix tables.

2.3 Intended Audience and Reading Suggestions

This document is intended for several stakeholders involved in the SmartReplenish lifecycle:

Project Manager: To monitor project scope, milestones, and resource allocation to ensure timely delivery.

System Architects & Developers: To understand the structural constraints and technical requirements (such as Kotlin and Azure SQL) for implementation.

Quality Assurance (QA) / Testers: To derive test cases from the iRTM and ensure every requirement is verifiable and testable.

Customer Stakeholders (ValuPrice): To verify that the specified requirements accurately reflect their original business intent and resolve previous incongruities

2.4 Product Scope

SmartReplenish is a mobile first inventory management solution developed for ValuPrice Pte Ltd. The system aims to automate household grocery tracking and streamline the replenishment process through direct retail integration.

The software will provide Household Users with the ability to scan item barcodes using mobile cameras, which then automatically populates item details by querying the GS1 Global Database. It monitors quantity levels and expiration dates, triggering automated alerts based on user-defined "High Risk" thresholds. Furthermore, the system integrates with the ValuPrice Master Database to provide real-time pricing and promotion data, ensuring that generated shopping lists are accurate and cost-effective. All data is synchronized via a cloud-native architecture on Azure Singapore using RESTful APIs

However, the scope of SmartReplenish is strictly limited to inventory management and replenishment planning. The system will not handle financial transactions or point-of-sale (POS) payment processing. Additionally, the software does not manage physical delivery logistics, fleet tracking, or the maintenance of the customer's mobile hardware. FreshTech Systems' responsibility is confined to the software application and its cloud deployment environment.

2.5 References

The following documents and standards have been used in the elicitation and analysis of the SmartReplenish requirements:

1. ValuPrice SmartReplenish Project Specification (PS) v1.0: The original elicitation document provided by the customer detailing their business needs and initial system vision.

2. IEEE 830-1998: IEEE Recommended Practice for Software Requirements Specifications used to organize this document.

3. Personal Data Protection Act 2012 (PDPA): The Singaporean regulatory framework governing data privacy, which dictates the encryption and residency requirements for this project.

4. ICT2113 Lecture Notes: The procedural and declarative knowledge framework for Requirements Engineering, including BCIC and iRTM methodologies.

5. GS1 General Specifications: The global standard for barcode identification used for the automated product catalog integration

3. Overall Description

3.1 Product Perspective

SmartReplenish is a new, self-contained Android-based mobile application specifically designed for the retail and household inventory market. While it operates as a standalone tool for the Household User, it is part of a larger ecosystem that integrates with external retail and global data systems to provide real-time value. The system serves as a bridge between the user's physical kitchen environment and ValuPrice Pte Ltd's retail database. As shown in the Requirements Tree in Section 1.4, the application's architecture is built on a RESTful architecture hosted on Microsoft Azure to ensure high availability and seamless data synchronization across multiple devices.

3.2 Product Functions

SmartReplenish provides a suite of automated tools to ensure efficient household inventory management and seamless grocery replenishment. Refer to Figure 1 in Appendix B for the Use Case Diagram.

3.2.1 Inventory Management

The system allows users to maintain an accurate digital record of their household goods.

- Barcode-Assisted Item Tracking: Enables rapid entry and identification of items using mobile camera scanning and the GS1 Global Database.

- Manual Entry: Provides a fallback for items without standard barcodes or for custom household goods.

- Expiry Tracking: Automatically monitors item perishability based on data logged during entry.

3.2.2 Automated Monitoring & Alerts

The system proactively monitors stock levels to prevent shortages.

- Stock Level Monitoring: Continuously tracks current quantities against user-defined "High Risk" thresholds.
- Low-Stock & Risk Alerts: Triggers automated mobile notifications when items fall below safety levels or are nearing expiration.

3.2.3 Replenishment & Retail Integration

The system streamlines the procurement of new supplies through direct retail connectivity.

- Auto-Shopping List Generation: Compiles depleted items into a structured shopping list for user review.
- Real-Time Price Sync: Integrates with the ValuPrice Master Database to ensure the shopping list reflects current pricing and active promotions.

3.2.4 User Account & Preference Management

Account & Security: Manages secure user profiles, password recovery protocols, and cross-device data synchronization.

3.3 User Classes and Characteristics

SmartReplenish is designed for a diverse group of users within the household and retail management ecosystem. Each user class has distinct levels of access and technical proficiency.

3.3.1 Household User (Primary Class)

The Household User is the primary beneficiary of the system, responsible for day-to-day inventory management.

- Characteristics: This user typically has a high frequency of use and moderate technical expertise with Android mobile devices.
- Responsibilities:
    - Scanning items for entry using the Barcode-assisted Tracking.
    - Setting "High Risk" thresholds for stock levels and expiration alerts.
    - Reviewing and managing the Auto-Shopping List.

3.3.2 System Administrator (Vendor-side)

Similar to the Account Administrators in the senior's SRS, this role manages the technical integrity of the platform.

- Characteristics: Highly technical users with full access to the backend cloud infrastructure on Microsoft Azure.
- Responsibilities:
  - Managing user account lifecycles and facilitating secure Password Resets.
  - Ensuring the availability of API documentation and maintaining PDPA compliance.

3.3.3 Retail Partner (ValuPrice Stakeholder)

This class represents the point of contact at the retail level for database integration.

- Characteristics: Professional users focused on data accuracy and business-to-business (B2B) synchronization.
- Responsibilities:
  - Ensuring the ValuPrice Master Database provides accurate pricing and promotion data for the Price Sync.

3.4 Operating Environment

SmartReplenish is designed to operate in a distributed mobile and cloud environment to ensure real-time inventory accessibility across various devices. The supported environments are:

Mobile Applications

- Android: Supports Android 6.0 (Marshmallow) and above to maintain compatibility with modern camera APIs and security protocols.
- Hardware: Physical smartphone device with a functional rear-facing camera for Barcode-assisted Item Tracking.

Cloud & Network Infrastructure

- Hosting: Backend environment hosted on Microsoft Azure Singapore.
- Connectivity: Requires stable internet access (WiFi/4G/5G) for Price Sync and Cloud Data Synchronization.

External Service Integration

- GS1 Database: Persistent API connection for automated item identification.

3.5 Design and Implementation Constraints

The development of SmartReplenish is governed by several technical and regulatory constraints to ensure system integrity and compliance:

- Regulatory Compliance: The system must adhere to the Personal Data Protection Act (PDPA) standards for Singapore, specifically regarding the encryption of user inventory data and account credentials.
- Architectural Style: The application must utilize a RESTful architecture to facilitate standardized communication between the mobile client and the Azure-hosted backend.
- Development Language: The mobile application is a native build using Java/XML to ensure high performance and reliable camera API integration.
- Data Persistence: All centralized inventory and user records must be maintained within an Azure SQL Database with automated recovery protocols.
- Interface Standards: Automated item identification is strictly constrained by the GS1 API protocols for barcode metadata retrieval.

3.6 User Documentation

To ensure successful system adoption and technical sustainability, FreshTech Systems will deliver the following documentation components alongside the software:

- User Manual and Tutorials: A comprehensive digital guide for Household Users, detailing the workflow for Barcode assisted Item Tracking and Auto-Shopping List Generation
- API Documentation: Technical specifications for the RESTful architecture and external integration protocols for the GS1 API.
- System Architecture Diagram: A high-level visual representation of the Azure-hosted backend and mobile client interaction to assist in future maintenance.
- Security & Recovery Protocol: Documentation outlining the data protection measures in place for PDPA compliance and the procedures for automated database recovery.
- Online Help System: An integrated FAQ within the mobile application to assist users with common tasks like Forgot Password protocols.

3.7 Assumptions and Dependencies

The successful implementation and operation of SmartReplenish are based on the following assumed factors and external dependencies:

- Third-Party API Stability: It is assumed that the GS1 Global Database and the ValuPrice Master Database will remain operational and accessible via their respective APIs.
- Hardware Capabilities: The system assumes that the Household User's mobile device has a functioning autofocus camera capable of scanning standard EAN/UPC barcodes.

- Cloud Service Availability: The project depends on the continuous availability of Microsoft Azure Singapore services for backend hosting and data persistence.
- User Connectivity: It is assumed that users will have intermittent or continuous internet access to facilitate the Cloud Data Synchronization and Price Sync.

4. External Interface Requirements

4.1 User Interfaces

The SmartReplenish UI follows Material Design standards and is optimized for one-hand use to support home inventory management. As detailed in Appendix B, the interface incorporates the following logical characteristics:

- Home Dashboard (Fig. 2.1): Displays real-time metrics for "Low Stock," "Total Items", and "To Order" with red "Urgent" badges for items below critical thresholds.
- Inventory Management (Fig. 2.2): Provides a filtered view of all household stock with "Critical" and "Low" status tags and a search interface for rapid item retrieval.
- Scanning Interface (Fig. 2.3): Features a camera viewfinder for 1D/2D barcodes with a manual text entry fallback for budget-device compatibility.
- Shopping List (Fig. 2.4): Automatically groups items by Store Category (BR-07) and incorporates Price Sync (BR-08) data to calculate total estimated costs.
- User Profile (Fig. 2.5): Manages account details, store location, and notification preferences, ensuring personalized alert thresholds.
- Interaction Design: Core navigation and a blue Floating Action Button (FAB) are positioned in the "thumb zone" for accessible multitasking.

4.2 Hardware Interfaces

The software shall interface with the following physical components of the mobile device:

- Camera Hardware: The application must interface with the device's camera to support 1D (EAN/UPC) and 2D (QR) barcode scanning.
- Network Interface: The app shall utilize the device's Wi-Fi or cellular radio to synchronize data with the Azure cloud backend
- Local Storage: The system shall interface with the device's internal flash storage to implement a local caching mechanism for offline viewing and editing.

- Push Notification Engine: The app must interface with the Android notification system to trigger low-stock and “consume soon” alerts even when the app is in the background.

4.3 Software Interfaces

The application integrates with several internal and external software components:

- Operating System: Android 6.0+
- Backend Services: A Cloud-Native RESTful architecture hosted on Microsoft Azure
- Database: Azure SQL (MSSQL) for all structured master data and user inventory logs
- External API (GS1): Integration with the GS1 Global Product Database API to automatically populate product names and images based on EAN/UPC codes
- ValuPrice Loyalty System: A secure interface to verify and link the user’s unique ValuPrice Loyalty ID during profile creation.
- API Protocols: All communication between the mobile client and backend shall be documented via OpenAPI

4.4 Communications Interfaces

Secure communication is a primary requirement for compliance and data integrity.

- Communication Protocol: This system shall use HTTPS for all data transmission between the mobile app and the RESTful backend
- Encryption Standards: Data in transit must be protected using TLS 1.2+ and data at rest using AES-256 to comply with Singapore PDPA regulations
- Authentication: The system shall utilize secure tokens (JWT) for session management and cryptographic, time-limited reset tokens delivered via registered email for account recovery.
- Synchronization Mechanism:
    - Background Sync: Price update must sync every 24 hours
    - Offline Conflict Resolution: The system shall preserve the server value and prompt the user to either accept the server value or re-apply the queued offline change

5. System Features

This section specifies the major functional capabilities of the SmartReplenish Android application, organized by feature groups that align with core user activities. Each feature is described with its priority, user-system interaction, and traceable functional requirements. The complete list can be found in Appendix D.

5.1 User Account Management

5.1.1 Description and Priority

This feature enables customers to create and manage personalized profiles linked to their ValuPrice Loyalty ID, ensuring seamless integration between in-store loyalty benefits and digital inventory management. The feature includes secure self-service account recovery to maximize user retention.

Priority: High

5.1.2 Stimulus/Response Sequences

1. User selects “Create Account”.
    a. System displays registration form requesting Loyalty ID, email, and password.
2. User enters valid Loyalty ID and submits.
    a. System validates ID against ValuPrice master database.
3. Validation succeeds.
    a. System creates profile, sends confirmation, displays dashboard.
4. User selects “Forgot Password”.
    a. System prompts for registered email address.
5. User submits email address.
    a. System generates time-limited cryptographic token, sends via email.
6. User clicks reset link within validity period
    a. System displays password reset form.
7. User submits new password.
    a. System updates credentials, invalidates token, confirms success.
8. User enters invalid Loyalty ID.
    a. System displays error: "Invalid Loyalty ID. Please check or visit store."

5.1.3 Functional Requirements

FR-01: Users shall create a profile linked to their ValuPrice Loyalty ID.
FR-09: The system shall implement secure password reset via time-limited cryptographic token delivered to registered email address.
FR-13: The system shall implement secure password reset protocol with unique time-limited tokens and industry-standard encryption.

5.2 Item Acquisition

5.2.1 Description and Priority

This feature provides dual-path entry through barcode scanning and manual search with mandatory categorization.

Priority: High

5.2.2 Stimulus/Response Sequences

1. User taps “Add Item”
    a. System presents choice: “Scan Barcode” or “Manual Entry”.

2. User selects “Scan Barcode”
    a. System activates camera, displays viewfinder with alignment guide.
3. Camera detects valid 1D barcode.
    a. System decodes data, queries product database.
4. Product found in database.
    a. System displays product name, requests category confirmation.
5. User selects “Manual Entry”.
    a. System displays search field and mandatory category selector
6. User selects category from taxonomy.
    a. System enables "Add to Inventory" button

5.2.3 Functional Requirements

FR-02: The system shall capture and decode 1D barcode data via device camera for item identification.
FR-03: The system shall enforce mandatory item categorization through user selection from a predefined taxonomy: Produce, Dairy & Chilled, Meat & Seafood, Bakery, Frozen, Pantry/Dry Goods, Beverages, Household, Personal Care, for all inventory entries.
FR-10: The system shall integrate device camera for 1D barcode capture and decoding.

5.3 Inventory Tracking & Alerts

5.3.1 Description and Priority

The feature enables real-time stock management through optimized one-hand interfaces, proactive low-stock notifications, and expiry date monitoring.

5.3.2 Stimulus/Response Sequences

1. User views inventory list.
    a. System displays items with current quantity, expiry indicators.
2. User taps “+” or “-” on item.
    a. System increments/decrements quantity, updates timestamp.
3. Quantity falls below user-defined threshold.
    a. System triggers push notification: “Low stock: [Item Name]”
4. User sets expiry date.
    a. System calculates alert date (72 hours prior).
5. Alert date reached.
    a. System sends push: “Consume soon: [Item Name] expires [Date]”.
6. Device loses connectivity.
    a. System switches to local cache mode (read-only).

5.3.3 Functional Requirements

FR-04: The system shall provide a simple +/- interface to update home stock.
FR-05: Users shall set thresholds that trigger replenishment notifications.
FR-06: The system shall monitor use-by dates and generate consumption alerts based on category-specific temporal thresholds.
FR-11: The system shall maintain local read-only cache of inventory data for offline viewing.

5.4 Shopping List Generation

5.4.1 Description and Priority

This feature automates shopping list creation based on inventory deficits end enriches lists with time pricing for budget planning.
Priority: High

5.4.2 Stimulus/Response Sequences

1. User taps “Generate Shopping List”.
    a. System queries all items below threshold
    b. System compiles list and group items by store category.
    c. System retrieves current prices and display price per item, calculates estimated total
2. Price sync fails (offline).
    a. System displays cache prices with “Prices may be outdated” warning.

5.4.3 Functional Requirements

FR-07: The system must compile a shopping list from items below thresholds.
FR-08: The system shall display real-time store prices via persistent connection, with cached fallback during connectivity loss.

5.5 Admin Dashboard Management

5.2.1 Description and Priority

This feature provides ValuPrice internal staff with secure web-based administrative capabilities to manage the master product catalogue and user accounts. Business ownership resides with the ValuPrice Marketing Team for product catalogue content management. Technical ownership transitions from FreshTech Systems (3-month warranty period) to ValuPrice IT Team for ongoing hosting, security, and maintenance on ValuPrice’s Azure subscription.
Priority: Medium

5.2.2 Stimulus/Response Sequences

1. Marketing Team member accesses dashboard URL.
    a. System presents Azure AD authentication screen.
2. Valid credentials with Marketing role confirmed.
    a. System displays "Product Catalogue" menu with search/filter options.
3. Marketing user edits product details (name, category, price, image).
    a. System validates changes, updates master database, logs modification.
4. Marketing user bulk-uploads product CSV.
    a. System validates format, processes rows, reports errors/success.
5. IT Team member accesses the dashboard with the Admin role.
    a. System displays additional "User Accounts" and "System Settings" menus.
6. IT user resets customer password.
    a. System generates a temporary password, sends it to the user, and forces a password change on login.
7. IT user views the security audit log.

   a. System displays PDPA-compliant access logs with an export option.
8. Unauthorized access attempt detected.
   a. System blocks IP, logs incident, and alerts ValuPrice IT Security.
9. The 3-month warranty period expires.
   a. System ownership transfers to ValuPrice IT; FreshTech moves to optional retainer.

5.2.3 Functional Requirements

FR-12: The system shall provide a secure web-based administrative interface for product catalogue and user account management.

6. Inceptive Requirements Traceability Matrix

6.1 Purpose

The Inceptive Requirements Traceability Matrix (iRTM) establishes bidirectional traceability between the Project Specification's business and technical requirements (BR/TR) and the system features functional requirements (FR). The document exists in a public repository for centralized access and version control and is attached as an external document (Annex 1).

6.2 Metadata and Matrix Components

The document specifies the Document Type for artifact classification, Project ID and Project name for initiative identification, Document date and Version for baseline tracking, and Assumption(s), Constraint(s), and Regulation(s) for contextual governance, collectively serving as the iRTM metadata.

The matrix specifies Serial for unique row identification.
The PS section captures:

- Category: Biz Ops or Tech classification
- ID: codification
- Description: paraphrased requirement
- Stakeholder: accountable role
- Precedence: Essential, Desirable, Acceptable ranking

The SRS section captures:

- Category: template taxonomy
- ID: codification
- Description: paraphrased requirement
- Owner: accountable role
- Ranking: MoSCoW prioritization

7. Other Nonfunctional Requirements

This section defines the nonfunctional requirements (NFRs) and business rules (BRs) that constrain how the system is developed and operated. These requirements address performance, safety, security, and overall software quality attributes that are necessary for a

reliable mobile user experience and a secure cloud-backed service. Each NFR is written to be measurable and verifiable through testing, inspection, or analysis. Where applicable, detailed verification steps and checklists are maintained in the project repository. Supporting documents are maintained in the SmartReplenish repository under:
https://github.com/Knight-RT/SmartReplenish/tree/main/01_Requirements
https://github.com/Knight-RT/SmartReplenish/tree/main/06_Testing_Evidence

7.1 Performance Requirements

The system shall provide responsive user interactions on the Android client and timely responses from cloud services to support key user flows such as viewing inventory, updating quantities, scanning barcodes, and synchronizing data. Performance requirements also ensure that network variability and background synchronization do not degrade usability.

The detailed performance NFRs and verification methods are consolidated in Appendix C (Figure C.1: Performance NFRs).

7.2 Safety Requirements

Safety requirements ensure the system behaves predictably and prevents avoidable harm to users during normal use and failure conditions. In this context, “harm” refers to outcomes such as accidental data loss, incorrect inventory states, misleading status information (e.g., sync freshness), and users making decisions based on outdated data.

The detailed safety NFRs and verification methods are consolidated in Appendix C (Figure C.2: Safety NFRs).

7.3 Security Requirements

The system shall protect user accounts and personal data through secure communication, strong authentication and authorization controls, and appropriate logging for accountability. Security requirements also support privacy obligations, and reduce common abuse patterns such as repeated login attempts.

The detailed security NFRs and verification methods are consolidated in Appendix C (Figure C.3: Security NFRs).

7.4 Software Quality Attributes

The software quality requirements are defined with reference to McCall’s Quality Model, which groups quality factors into Product Operation, Product Revision, and Product Transition. The NFRs below specify measurable quality targets to ensure the system is usable and dependable in production, maintainable during iterative development, and compatible across target platforms and integrations.

The detailed McCall-mapped quality attribute NFRs and verification methods are consolidated in Appendix C (Figure C.4: McCall Quality Attribute NFRs).

7.5 Business Rules

Business rules define non-negotiable constraints derived from operational needs and governance. These rules guide system behavior across inventory management, pricing display, offline operation, and administrative controls. Business rules are distinct from functional requirements in that they constrain how features operate and what outcomes are permitted.

The complete business rules catalogue is consolidated in Appendix C (Figure C.5: Business Rules).

8. Other Requirements

This section specifies additional system requirements that are not covered in earlier non-functional sections. These requirements focus primarily on data management, regulatory compliance, and system extensibility to ensure the SmartReplenish application remains scalable, maintainable, and compliant with industry standards.

8.1 Data Requirements

The SmartReplenish system requires a structured relational database to store application data, including user accounts, product catalogues, inventory records, shopping lists, and alert notifications.

The database schema is designed to support the core functional features of the system, including barcode-based product acquisition, inventory monitoring, automated shopping list generation, and administrative product catalogue management.

The Entity Relationship Diagram (ERD) illustrating the relationships between these entities is provided as Figure 3, in Appendix B. The detailed Data Dictionary describing each entity and attribute is also provided in Appendix B, Figure 4 to Figure 10.

9. Change Mitigation Strategies

9.1 Anticipatory Contingency Requirements

ID: CM-ACR-01:
Description: The system shall provide a pre-defined migration path from Azure SQL Standard to Azure SQL Hyperscale tier, triggered automatically when performance monitoring detects a sustained 80% DTU or storage limit over a 1-hour window.

9.2 Open Issues

ID: CM-OI-01: Azure Environment Provisioning and Access

Description: While ValuPrice has confirmed hosting on their Azure Subscription, the provisioning of credentials and environment access remains pending. The final deployment and production testing are currently blocked
Impact: Failure to secure production access prevents environment testing, CI/CD pipeline establishment, and final deployment.
Mitigation: FreshTech will maintain a shadow production environment within its internal sandbox.
Kill Date: 4 May 2026. If access is not provided by this date, the project will transition to a handover model where FreshTech delivers a deployment package for ValuPrice to install internally, and production testing will be formally removed from FreshTech's responsibility.

9.3 High Risk Change Control

ID: CM-HRCC-01
Reference: FR-03 Predefined Taxonomy for Categorization
Risk: The predefined 9 categories are embedded into the database schema.
Change Control: Any request to re-integrate the GS1 database or modify the taxonomy must be submitted via a Formal Change Request (CR) to the FreshTech Business Analyst.
Acceptance Condition: The CR will be accepted only if it includes a confirmed budget allocation for GS1 licensing and technical validation of the GS1 API's regional data accuracy in Singapore.
Priority: High
Value Assessment: Manual entry has fulfilled the business value of the Shopping List Generation feature. The integration of GS1 is considered an enhancement to provide convenience.
Technical Impact: A rollback on both the Item Acquisition feature UI and a reconfiguration of the architecture to handle the external API.
Negotiation Strategy: FreshTech will hold a favourable position, providing the manual fallback as a sign of goodwill by keeping the Data Schema GS1-Ready as a compromise.
Kill Date: May 1, 2026. If a GS1 license is not procured, the GS1-Ready fields will be deprecated from the existing schema to optimize database performance, and manual entry will become the permanent system of record.

10. Sign Off

This Software Requirements Specification (SRS) document records the comprehensive requirements definition work completed during the Specifying stage for the SmartReplenish Android Mobile Application project. It details functional and non-functional requirements, system interfaces, constraints, and acceptance criteria derived from the prior BCIC analysis phase.

By signing below, the approvers confirm that they have reviewed this SRS and agree that:

- The requirements accurately represent the agreed-upon stakeholder needs, business objectives, and system scope as validated during the Specifying stage
- The documented functional requirements, non-functional requirements, and acceptance criteria are complete, consistent, and traceable to the source analysis outcomes
- The technical constraints, assumptions, dependencies, and identified risks are valid and acknowledged
- The requirements baseline established in this SRS is approved for proceeding to design, development, and verification activities

This sign-off formally establishes the SRS baseline and authorizes downstream engineering work. Any material changes, additions, or modifications to these baselined requirements after sign-off must be recorded as formal Change Requests and managed through the project's established change control process (CCB). No deviations from this baseline shall be implemented without prior approval.

Role / Organisation	Name	Signature	Date
Project Lead, ValuPrice	David Chen		
Senior Project Manager, FreshTech Systems	Sarah Lim		

FreshTech Systems

TechPoint Tower, 88 Innovation Drive,
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Appendix A: Glossary

Data Transaction Unit (DTU)
A unit of measure that represents a mix of measures of CPU, memory, reads, and writes.

Change Request (CR)
A formal proposal to modify a system, process, or project and is used in software development.

Software Requirements Specification (SRS)
A document that defines what a software system must do (functional requirements) and the constraints/qualities it must satisfy (nonfunctional requirements).

Functional Requirement (FR)
A requirement that describes a specific system behavior or feature the system must provide to users or other systems.

Nonfunctional Requirement (NFR)
A requirement that defines quality attributes or constraints on the system (e.g., performance, security, usability) rather than specific features.

Business Rule (BR)
A non-negotiable constraint derived from business operations or governance that limits or guides how system features may behave.

Inceptive Requirements Traceability Matrix (iRTM)
A traceability table that links each requirement to its source, related requirements, and supporting artifacts to maintain end-to-end traceability.

Traceability
The ability to link requirements to their origins and to downstream artifacts such as design, code, and test cases.

Requirement Codification
A consistent ID naming convention (e.g., FR-xx, NFR-SEC-xx) used to uniquely identify and reference requirements across documents and artifacts.

Use Case (UC)
A description of interactions between an actor and the system to achieve a goal, often used to capture functional behavior and system scope.

Actor
A role (user or external system) that interacts with the system to achieve a goal.

MoSCoW Prioritization
A prioritization technique categorizing requirements into Must have, Should have, Could have, and Won’t have.

Operating Environment
The hardware, software, and platform conditions where the system is expected to run (e.g., Android version, cloud hosting).

Offline Mode
A mode where the client application operates with limited functionality due to lack of network connectivity.

Offline Queue
A local list of changes made while offline that are stored and synchronized when connectivity returns.

Synchronization (Sync)
The process of aligning client-side data with server-side data to achieve a consistent state.

Synchronization Conflict
A conflict that occurs when the same record is modified both locally and on the server before synchronization completes.

Conflict Resolution
The rules or process used to decide what to do when a synchronization conflict occurs.

Personal Data Protection Act (PDPA)
Singapore's law governing the collection, use, disclosure, and retention of personal data.

Role-Based Access Control (RBAC)
An authorization method where system permissions are assigned based on user roles.

JSON Web Token (JWT)
A signed token used to represent an authenticated session between client and server.

Transport Layer Security (TLS)
A security protocol that encrypts network traffic (commonly used with HTTPS).

Advanced Encryption Standard (AES-256)
A standard encryption algorithm commonly used to protect stored data.

GS1
An international standards organization that defines product identification standards such as barcodes.

EAN/UPC (Barcode Standards)
Barcode formats used to uniquely identify retail products.

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Appendix B: Analysis Models

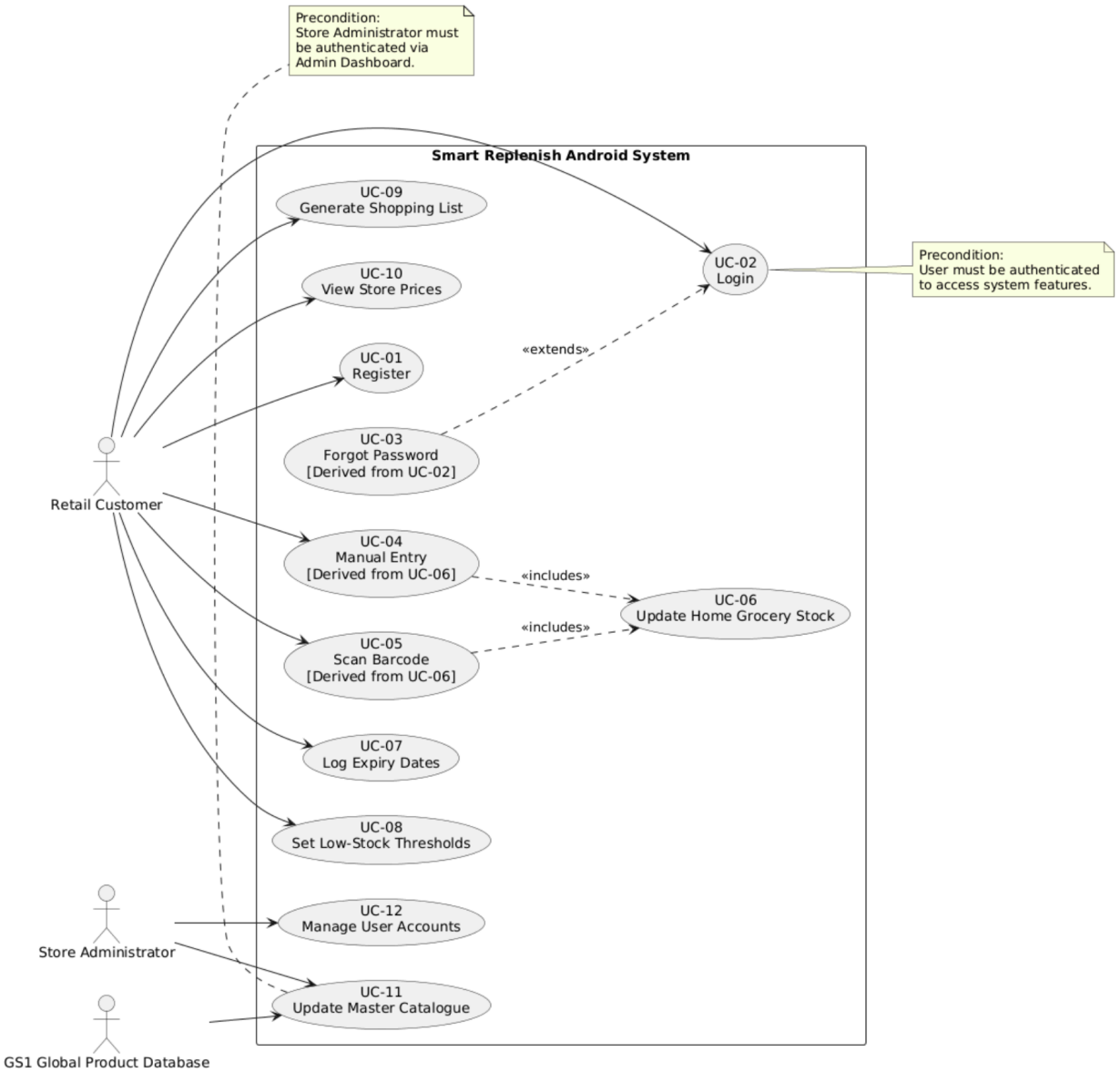

Figure 1: Use Case Diagram

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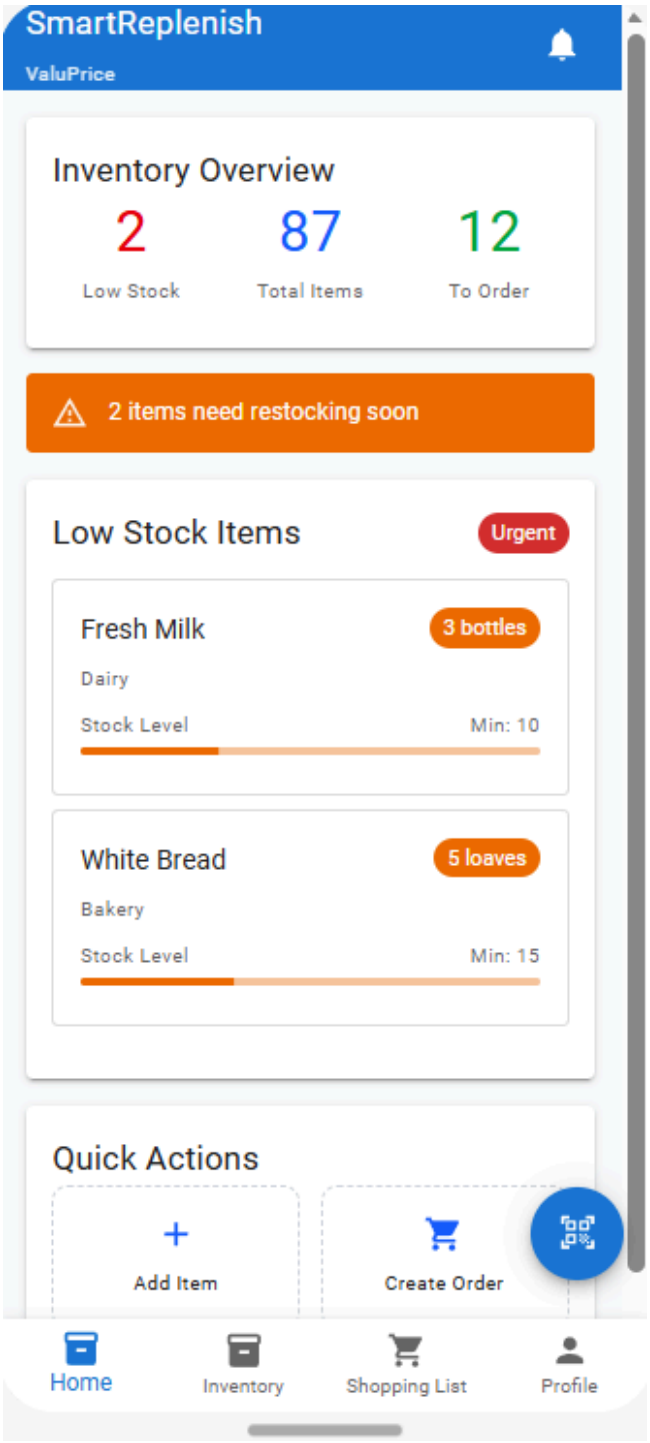

Figure 2.1: High-Fidelity Wireframe of the SmartReplenish Home Dashboard

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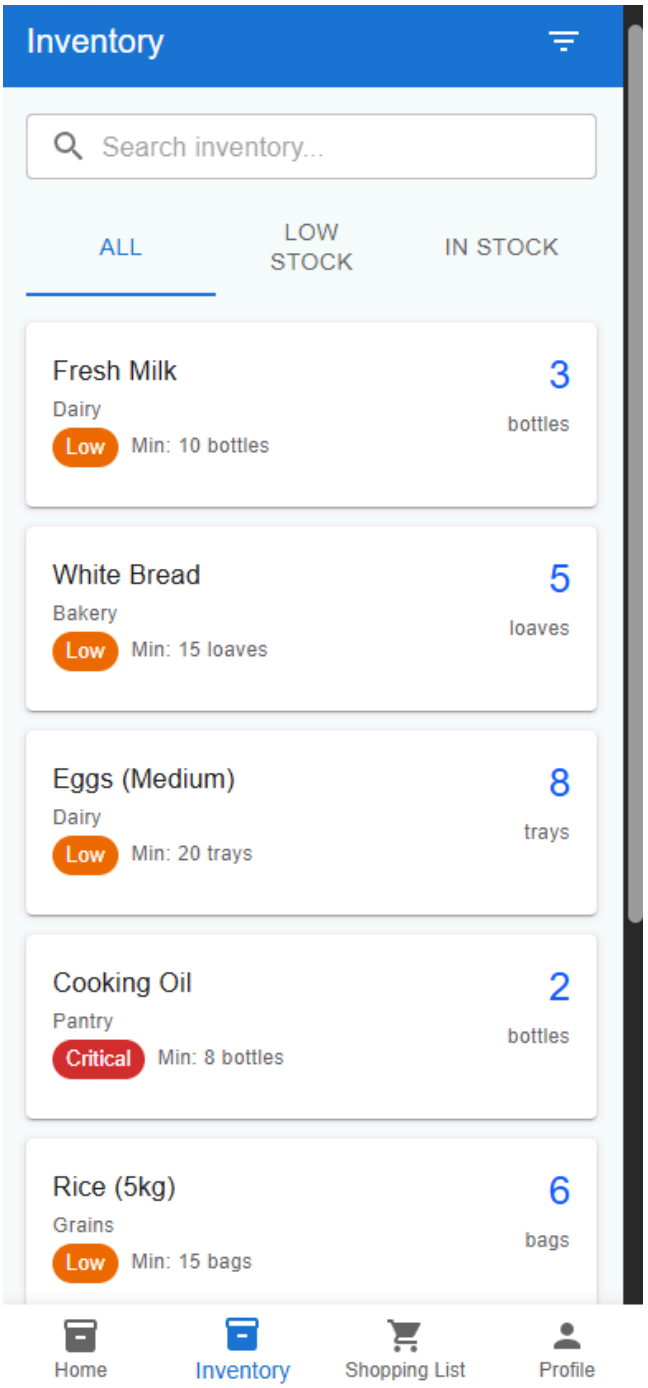

Figure 2.2: High-Fidelity Wireframe of the SmartReplenish Inventory Management Screen

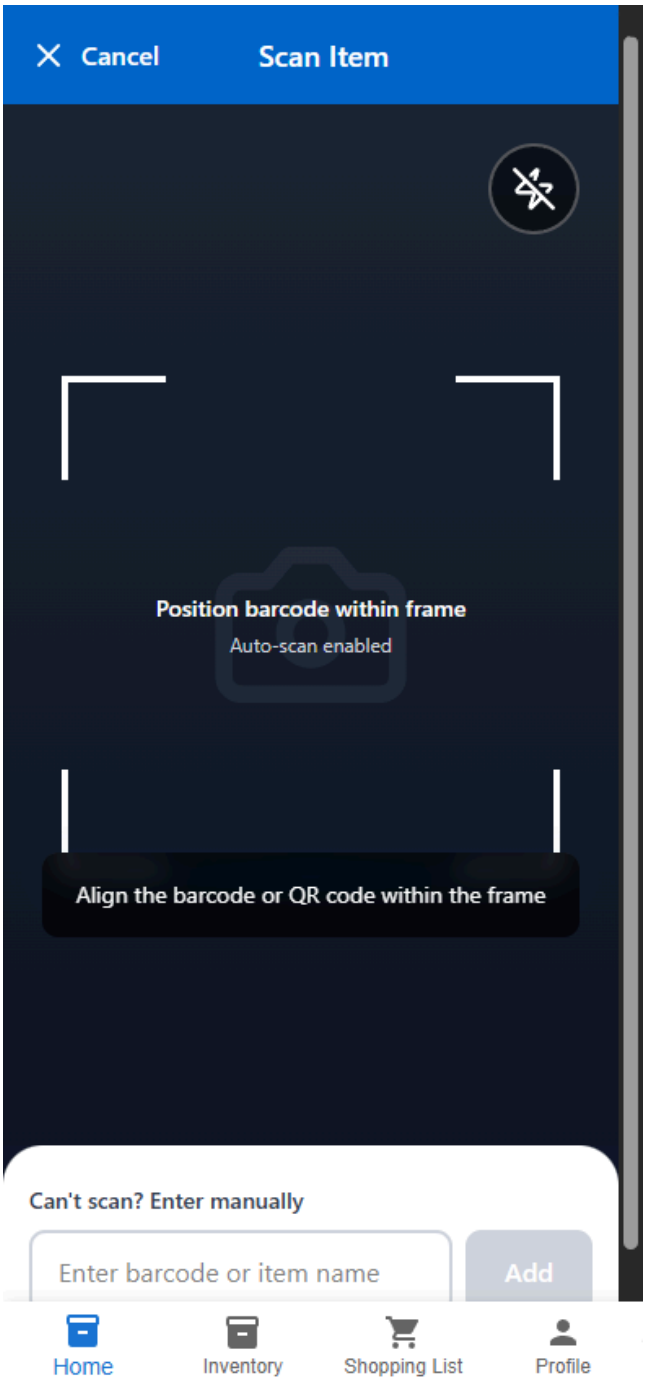

Figure 2.3: High-Fidelity Wireframe of the SmartReplenish Barcode Scanning Interface

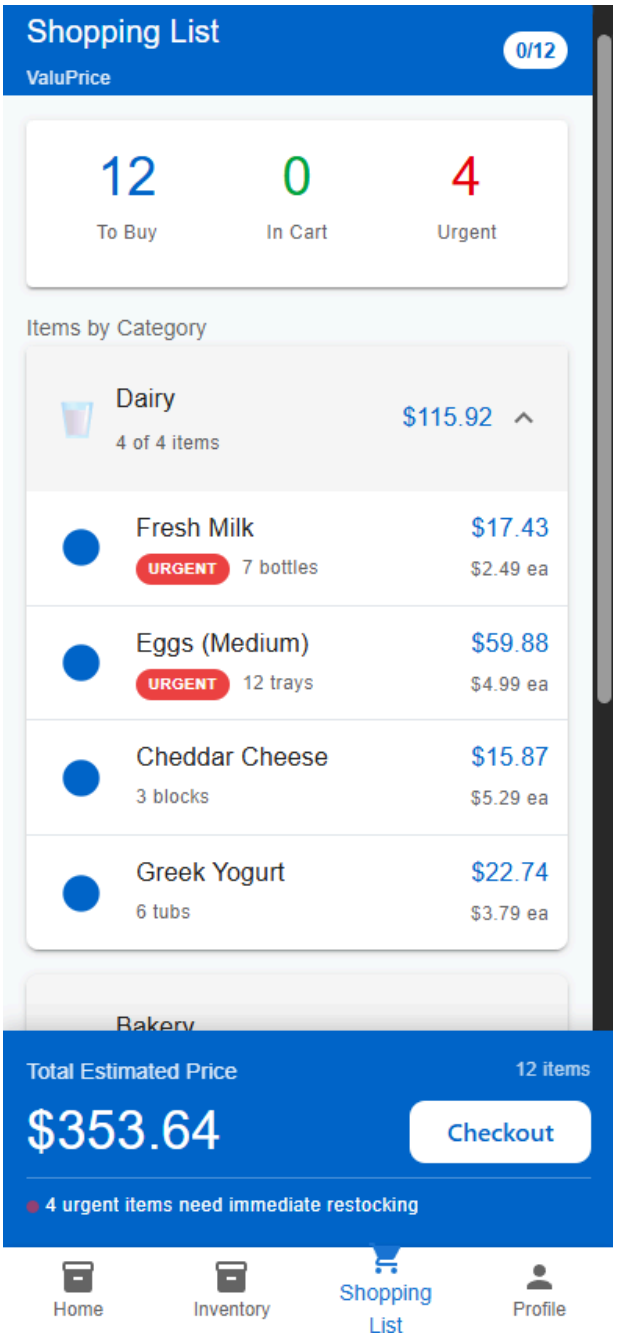

Figure 2.4: High-Fidelity Wireframe of the SmartReplenish Automated Shopping List

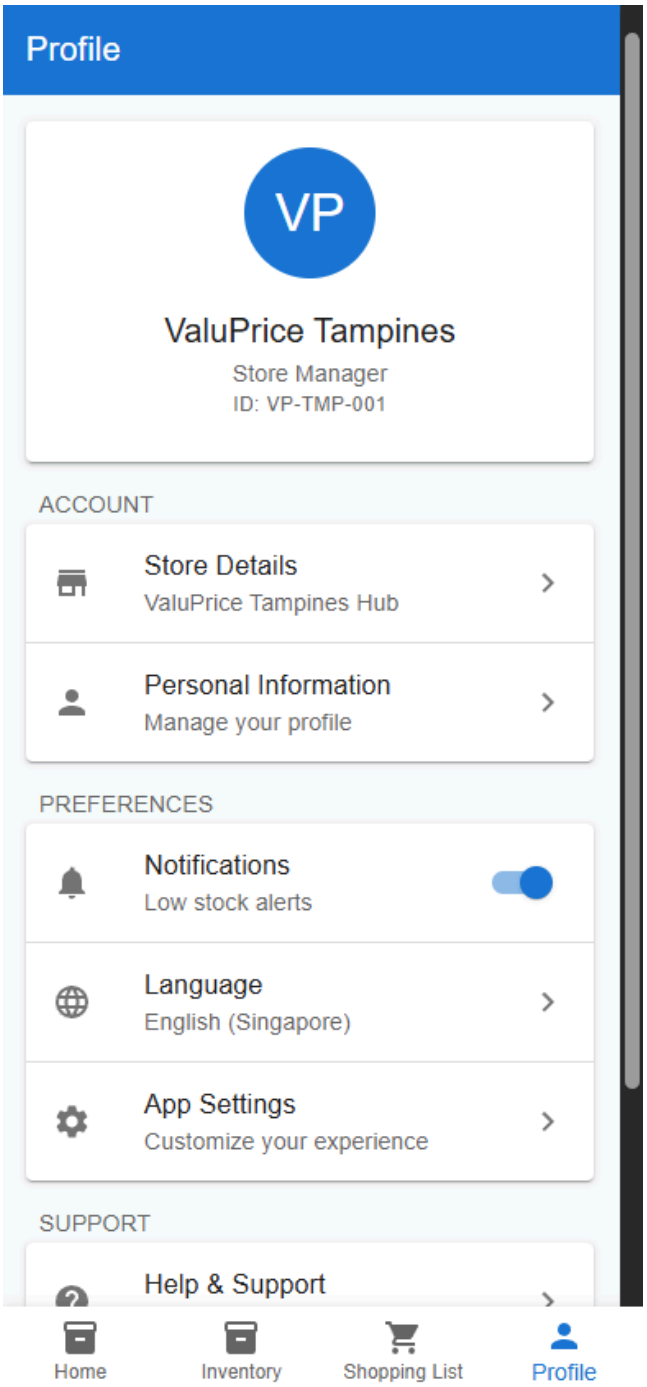

Figure 2.5: High-Fidelity Wireframe of the SmartReplenish User Profile & Settings

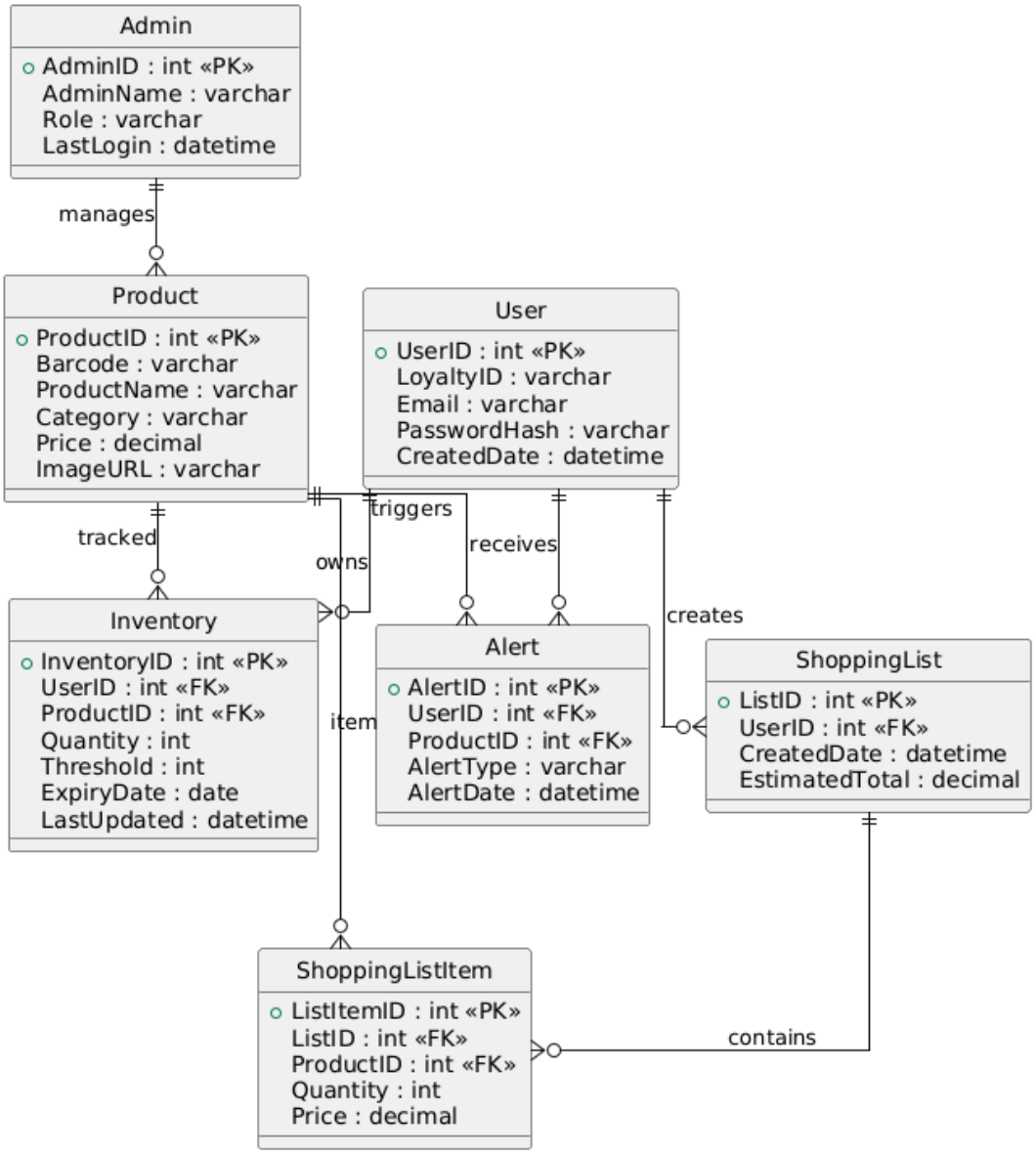

Figure 3. Entity Relationship Diagram

Field	Type	Description
UserId	INT (PK)	Unique identifier for each user
LoyaltyId	VARCHAR (50)	Loyalty card identifier
Email	VARCHAR (100)	Registered email address
PasswordHash	VARCHAR (255)	Encrypted user password
CreatedDate	DATETIME	Account creation timestamp

Figure 4. User Table Data Dictionary

Field	Type	Description
ProductId	INT (PK)	Unique product identifier
Barcode	VARCHAR (50)	Product Barcode
ProductName	VARCHAR (150)	Name of product
Category	VARCHAR (50)	Product category
Price	DECIMAL	Current store price
ImageURL	VARCHAR (255)	Product image link

Figure 5. Product Table Data Dictionary

Field	Type	Description
InventoryId	INT (PK)	Unique inventory record
UserId	INT (FK)	Owner of the inventory
ProductId	INT (FK)	Product reference
Quantity	INT	Current stock quantity
Threshold	INT	Minimum stock level
ExpiryDate	DATE	Product expiry date
LastUpdated	DATETIME	Last update timestamp

Figure 6. Inventory Table Data Dictionary

Field	Type	Description
ListId	INT (PK)	Unique shopping list identifier
UserId	INT (FK)	Owner of the list
CreatedDate	DATETIME	Date the list was generated
EstimatedTotal	DECIMAL	Estimated total cost

Figure 7. ShoppingList Table Data Dictionary

Field	Type	Description
ListItemId	INT (PK)	Unique item identifier
ListId	INT (FK)	Associated shopping list
ProductId	INT (FK)	Product reference
Quantity	INT	Quantity to purchase
Price	DECIMAL	Current item price

Figure 8. ShoppingListItem Table Data Dictionary

Field	Type	Description
AlertId	INT (PK)	Unique alert identifier
UserId	INT (FK)	User receiving alert
ProductId	INT (FK)	Product triggering alert
AlertType	VARCHAR (50)	Low Stock / Expiry Warning
AlertDate	DATETIME	Date alert generated

Figure 9. Alert Table Data Dictionary

Field	Type	Description
AdminId	INT (PK)	Unique admin identifier
AdminName	VARCHAR (100)	Name of admin
Role	VARCHAR (50)	Marketing / IT admin
LastLogin	DATETIME	Last system access

Figure 10. Admin Table Data Dictionary

Appendix C: Nonfunctional Requirements and Business Rules

Figure C.1 - Performance NFRs

ID	Name	Requirements	Verification
NFR-PR-01	Cold start time	The system shall reach the Home screen within 4 seconds on a reference Android 6.0+ device under normal conditions.	Test
NFR-PR-02	Warm resume time	The system shall resume from the background within 2 seconds.	Test
NFR-PR-03	Inventory list load	The system shall load the inventory list view within 2 seconds for up to 300 items on a reference device.	Test
NFR-PR-04	Item update responsiveness	The system shall reflect quantity or expiry updates on-screen within 1 second after a successful save action.	Test
NFR-PR-05	Scan-to-result latency	The system shall display the item identification result within 2 seconds when online following a barcode scan.	Test
NFR-PR-06	API response time	The system shall respond to at least 95% of backend API requests within 1.5 seconds under normal load.	Test / Log
NFR-PR-07	Sync after reconnect	The system shall complete background synchronization within 60 seconds after connectivity is restored for a typical user dataset.	Test
NFR-PR-08	Degraded network usability	The system shall remain responsive under poor network conditions and display progress indicators instead of becoming unresponsive.	Test
NFR-PR-09	Non-blocking price refresh	The system shall not block core user actions during price updates; if price retrieval exceeds 3 seconds, the system shall display cached values with a freshness indicator.	Test / Inspection
NFR-PR-10	Resource stability	The system shall not exhibit runaway memory growth during continuous use	Test (profiler)

		for 15 minutes on a reference device.	

Figure C.2 - Safety NFRs

ID	Name	Requirements	Verification
NFR-SAF-01	Confirmation for destructive actions	The system shall require user confirmation before destructive actions (e.g., deleting an item, clearing inventory).	Test / Inspection
NFR-SAF-02	Undo where feasible	The system shall provide an “Undo” option for item deletion actions within 10 seconds after completion, where feasible.	Test
NFR-SAF-03	Input validation	The system shall reject invalid inputs (e.g., negative quantities, invalid date formats) and provide an error message that guides correction.	Test
NFR-SAF-04	Consistent “last updated” display	The system shall display a clear “Last synced at …” timestamp on relevant screens when offline or when synchronization fails.	Test
NFR-SAF-05	Safe degraded mode	The system shall degrade safely when backend services are unreachable by preventing actions that could create inconsistent states and informing the user of the limitation.	Test
NFR-SAF-06	Non-misleading expiry messaging	messaging The system shall present expiry alerts and recommendations as guidance and shall not claim official certification of food safety.	Inspection

Figure C.3 - Security NFRs

ID	Name	Requirements	Verification
NFR-SEC-01	Secure transport	The system shall ensure all client-server communication uses HTTPS with TLS 1.2 or above.	Test / Inspection
NFR-SEC-02	Encryption at rest	The system shall encrypt sensitive data stored at rest using AES-256 or an equivalent managed encryption mechanism on the cloud platform.	Inspection
NFR-SEC-03	Token-based authentication	The system shall use token-based authentication with token expiry, and the system shall reject requests authorized by expired tokens.	Test
NFR-SEC-04	Secure token storage	The system shall store authentication tokens using secure device storage mechanisms and the system shall not log tokens in plaintext.	Test / Inspection
NFR-SEC-05	Password reset expiry	The system shall issue password reset tokens/links that are single-use and expire within 24 hours.	Test
NFR-SEC-06	Role-based access control	The system shall enforce role-based access control (RBAC) for administrative functions to restrict privileged actions to authorized roles.	Test
NFR-SEC-07	Audit logging	The system shall log security-relevant events (e.g., login, failed login, password reset requests, catalogue edits, role changes) with timestamp and user identifier.	Inspection
NFR-SEC-08	Rate limiting	The system shall rate-limit login and password reset requests (e.g., 5 attempts per 10 minutes, set by team) to mitigate abuse.	Test
NFR-SEC-09	Privacy minimization	The system shall collect and store only personal data necessary for defined business purposes and the system shall not store unnecessary identifiers.	Inspection
NFR-SEC-10	Data request support	The system shall provide an administrative process to support user	Test / Inspection

		requests for access, correction, or deletion of personal data subject to policy.	

Figure C.4 - McCall Quality Attribute NFRs

ID	McCall Factor	Requirement	Verification
NFR-QA-01	Usability	The system shall enable users to complete core tasks (view inventory, update quantity, add item) with no more than 5 primary taps per task under normal conditions.	Test
NFR-QA-02	Correctness	The system shall validate input values to prevent invalid inventory states (e.g., negative quantity, invalid dates) and provide an actionable error message.	Test
NFR-QA-03	Reliability	The system shall ensure confirmed inventory updates persist after application restart and are not lost due to app closure.	Test
NFR-QA-04	Efficiency	The system shall meet the performance targets specified in Section 6.1 (e.g., inventory list load time, scan latency, and API response time).	Test / Log
NFR-QA-05	Integrity	The system shall restrict access to administrative functions to authorized roles and log security-relevant events as defined in Section 6.3.	Test / Inspection
NFR-QA-06	Maintainability	The system shall maintain up-to-date API documentation (e.g., OpenAPI/Swagger) whenever endpoints or payloads change.	Inspection
NFR-QA-07	Testability	The system shall separate core business logic from UI concerns such that unit tests can validate business rules without requiring UI execution.	Test / Inspection
NFR-QA-08	Flexibility	The system shall support changes to product categorization or pricing rules without requiring changes across unrelated modules.	Inspection
NFR-QA-09	Portability	The system shall support Android 6.0+ and degrade gracefully on low-end devices (no crashes; reduced visual effects permitted).	Test
NFR-QA-10	Interoperability	The system shall provide a fallback path (e.g., manual entry) when external lookup services fail, without blocking core flows.	Test
NFR-QA-11	Reusability	The system shall implement shared validation and formatting components (e.g., date validation, quantity rules, price display formatting) as reusable modules that are consistently applied across	Inspection

		screens.	

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Figure C.5 - Business Rules

ID	Rule	Rationale	Related Use Case / Feature	Owner / Source
BR-01	The system shall ensure each inventory item belongs to exactly one category.	Enables consistent filtering, analytics, and alerts.	Add/Update Item	Product Owner / BA
BR-02	The system shall prevent item quantity from being negative.	Prevents invalid inventory states.	Update Quantity	BA
BR-03	The system shall prevent an expiry date from being earlier than the item's creation/add date.	Prevents invalid date records and misleading alerts.	Add/Update Item	BA
BR-04	The system shall allow users to manually add an item and mark it as "unverified" (or equivalent label) when barcode lookup fails.	Ensures users are not blocked by lookup failures.	Scan/Add Item	BA
BR-05	The system shall allow users to create and update inventory items while offline by storing changes locally and queuing them for synchronization when connectivity is restored.	Ensures predictable behavior and prevents data conflicts.	Offline Usage	BA / Tech Lead
BR-06	The system shall display a "last updated" timestamp when showing cached price information.	Prevents misleading representation of real-time pricing.	View Item / Price Display	BA
BR-07	The system shall enforce that a loyalty identifier is unique to one user account at a time.	Prevents account duplication and identity conflicts.	Registration/Login	BA
BR-08	The system shall restrict administrative catalogue updates to authorized roles (e.g., Marketing).	Ensures governance and prevents unauthorized changes.	Admin Dashboard	Business Stakeholder
BR-09	The system shall log security-sensitive administrative actions (e.g., role changes, account actions).	Supports accountability and audits.	Admin Dashboard	IT / Admin Stakeholder

BR-10	The system shall enforce personal data retention and deletion in accordance with the project's defined policy.	Aligns with privacy compliance expectations.	Account/Data Management	BA / Compliance
BR-11	The system shall persist queued offline changes across app restarts until synchronization succeeds or the user discards them.	Prevents loss of user updates made offline and ensures eventual consistency.	Offline Usage	BA / Tech Lead
BR-12	The system shall synchronize queued offline changes in chronological order (first-in, first-out) when connectivity is restored.	Ensures predictable outcomes and prevents unintended overwrites from out-of-order updates.	Offline Usage	Tech Lead
BR-13	The system shall detect a synchronization conflict when the same inventory record is modified both locally (offline) and on the server after the user's last successful sync.	Prevents silent overwriting of newer data and enables controlled conflict handling.	Offline Usage / Sync	Tech lead
BR-14	The system shall resolve synchronization conflicts by preserving the server value and prompting the user to either accept the server value or re-apply their queued offline change.	Prevents accidental overwriting of newer server data while allowing the user to make an informed decision.	Offline Usage / Sync	BA / Tech Lead
BR-15	The system shall display the synchronization status of offline changes (e.g., Pending Sync, Syncing, Synced, Conflict/Failed) to the user.	Improves transparency and prevents users from relying on stale or unsynchronized information.	Offline Usage / Sync Status	BA

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Appendix D: Functional Requirements

Functional Requirement	MosCow
Users shall create a profile linked to their ValuPrice Loyalty ID.	Must-have
The system shall capture and decode 1D barcode data via device camera for item identification.	Must-have
The system shall enforce mandatory item categorization through user selection from a predefined taxonomy: Produce, Dairy & Chilled, Meat & Seafood, Bakery, Frozen, Pantry/Dry Goods, Beverages, Household, Personal Care, for all inventory entries.	Must-have
The system shall provide a simple +/- interface to update home stock.	Must-have
Users shall set thresholds that trigger replenishment notifications.	Should-have
The system shall monitor use-by dates and generate consumption alerts based on category-specific temporal thresholds.	Should-have
The system must compile a shopping list from items below thresholds.	Must-have
The system shall display real-time store prices via persistent connection, with cached fallback during connectivity loss.	Should-have
The system shall implement secure password reset via time-limited cryptographic token delivered to registered email address.	Must-have
The system shall integrate device camera for 1D barcode capture and decoding.	Must-have
The system shall maintain local read-only cache of inventory data for offline viewing.	Must-have
The system shall provide secure web-based administrative interface for product catalogue and user account management.	Could-have
The system shall implement secure password reset protocol with unique time-limited tokens and industry-standard encryption.	Must-Have

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Annex 1: Inceptive Requirements Traceability Matrix

File Name: 1376T_SmartReplenish_InceptiveRequirementsTraceabilityMatrix

Github Link: https://github.com/Knight-RT/SmartReplenish/tree/main/04_Traceability